Q4: Machine learning

- Due 16 Jan at 12:30
- Points 5
- Questions 5
- Available after 20 Nov 2025 at 12:00
- Time limit 60 Minutes
- Allowed attempts 5

This quiz is no longer available as the course has been concluded.

Attempt history

	Attempt	Time	Score
LATEST	Attempt 1	3 minutes	5 out of 5

ⓘ Correct answers are hidden.

Score for this attempt: 5 out of 5
Submitted 30 Nov 2025 at 19:30
This attempt took 3 minutes.

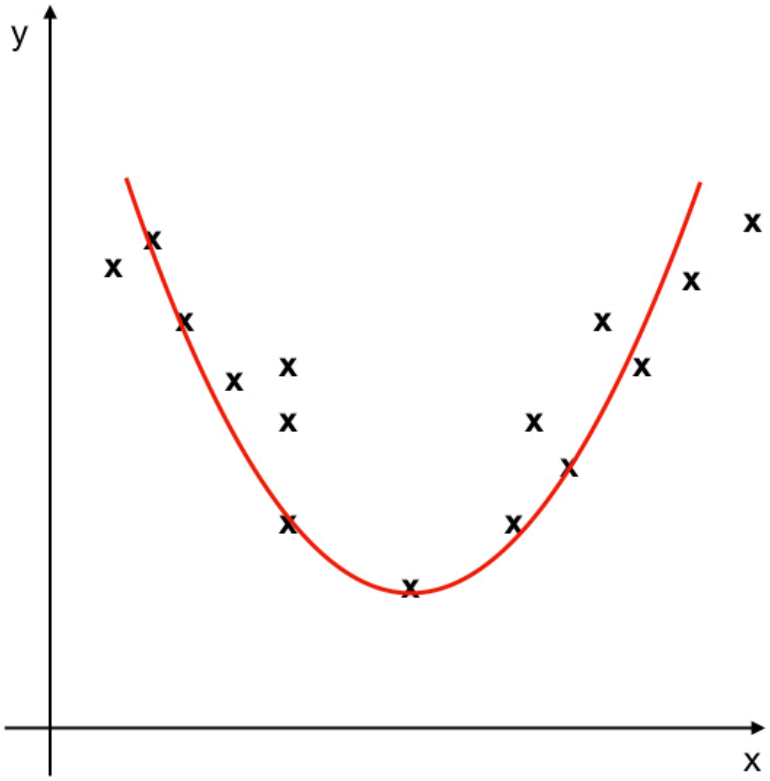
Question 1
1 / 1 pts
Mark all cases where supervised learning should be applied as opposed to unsupervised learning:

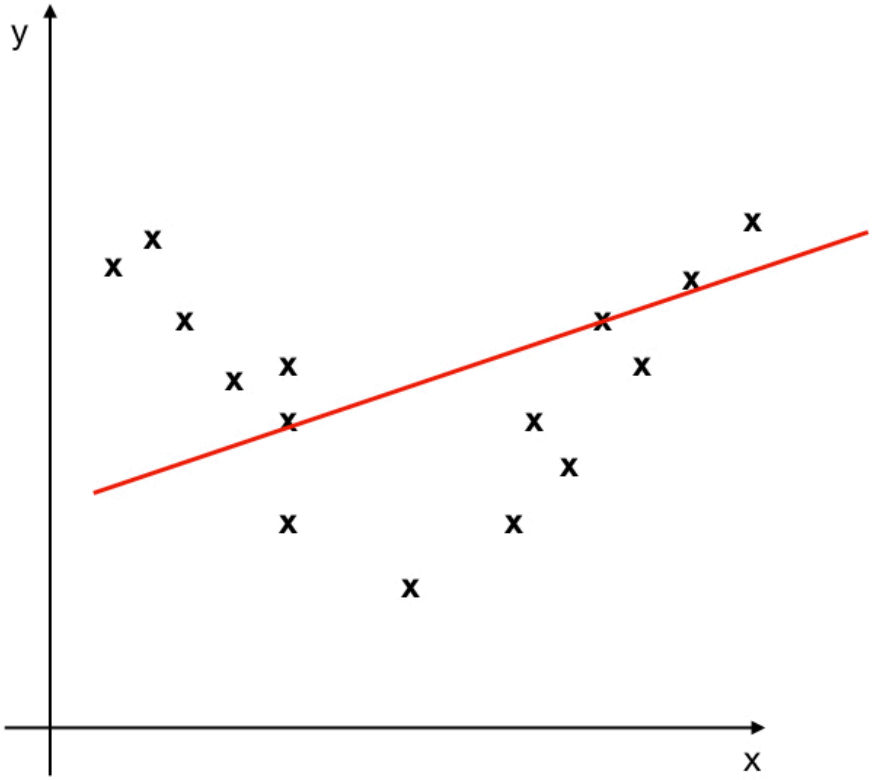
- [ ] You are given a database of speech samples. You would like to split them into clusters that contain samples from the same person.
- [x] You are given a database of text samples and for each of them, you know which language it is written in. You want to predict the language of a new text.
- [x] You are given a database of photos of people that are smiling and a database of pictures of people that are not smiling. You would like to recognize whether a person on a new picture is smiling or not.
- [ ] A set of people gave you information about their interests and hobbies. To each of them, you would like to suggest a few people they might like to meet.

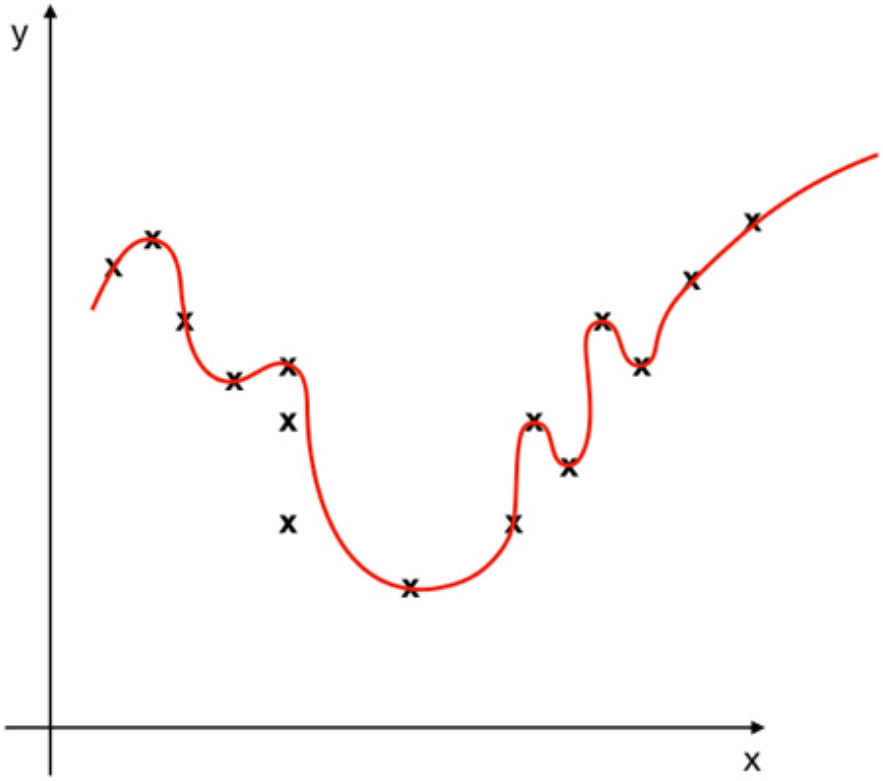
Question 2

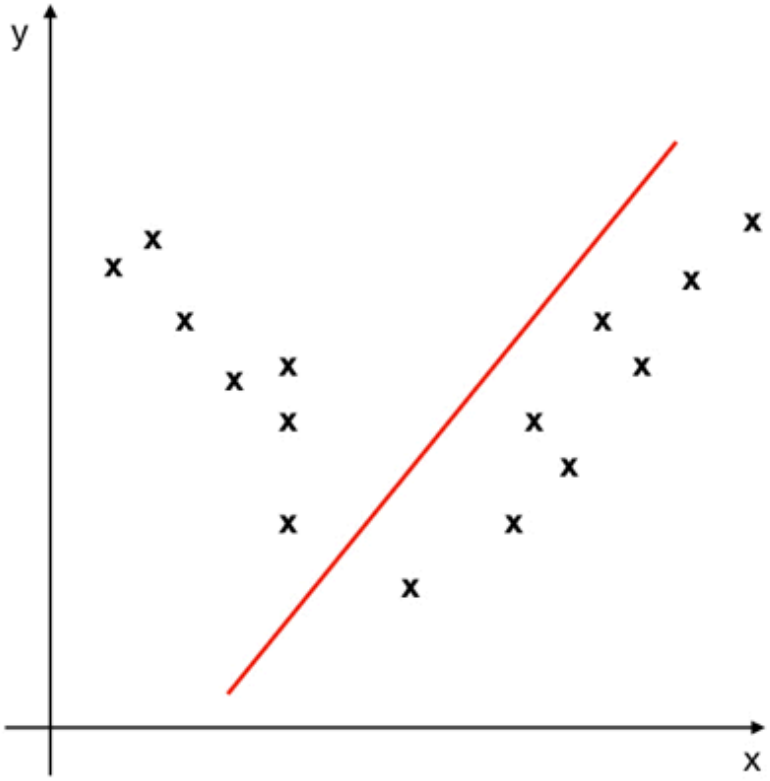
1 / 1 pts

Which out of the following functions is the most suitable one to be a model for the given data in regression learning?




y
x


y
x



Question 3

1 / 1 pts

The model in linear regression is $f(x) = wx + b$. Given the following observations, what are the best values when using least squares optimization criteria?

- w = 2.0, b = 2.7
- w = -2.0, b = -2.7
- w = -2.0, b = 2.7
- w = 2.0, b = -2.7

Question 4

1 / 1 pts

Which of the following approaches is best when deciding what model to use, e.g. when deciding what polynomial degree to use in regression?

- Train different models on all data, pick the one with lowest error.
- First, split the data into training and validation set. Train the models on the train data, and pick the one with lowest error on the validation set.
- Always pick a simple model (low polynomial degree), since otherwise we might overfit to the training data.
- Always pick a flexible model (high degree polynomial), since it can fit any data.

Question 5

1 / 1 pts

In supervised learning, we can add a regularization term to the goodness-of-fit function. Why do we do this? Choose the most correct answer.

- It removes the need to manually set the polynomial degree.
- It makes the optimisation easier, i.e. we find the solution faster.
- It changes the task to classification, which is easier.
- Regularisation is a way to avoid over-fitting.

Quiz score: 5 out of 5